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Support Vector Machines for Classification



Session Overview

Understanding Decision Boundaries



Objective

- Relate discriminant function theory to real-world data visuals

Focus Areas

- Gaussian contours
- Linear vs nonlinear boundaries
- Real dataset classification

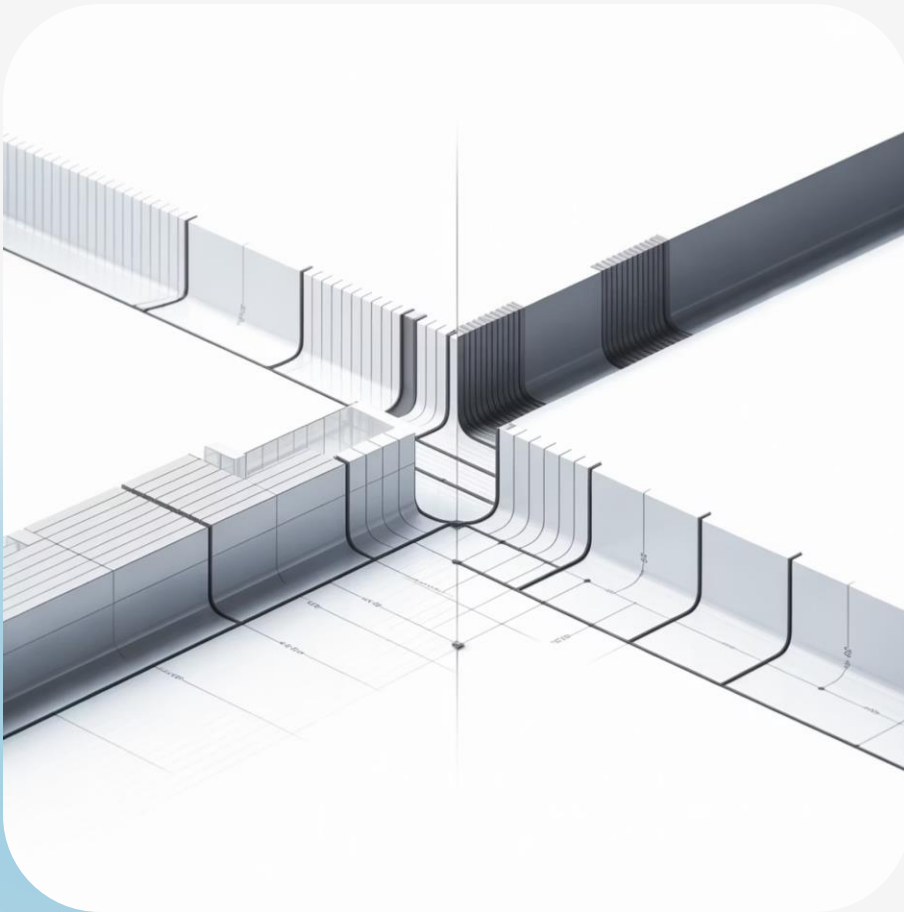
Key Outcome

- Understand how algorithms form decision boundaries

Explore Perceptron, Logistic Regression, and GMM using real dataset demos

Concept Recap

Core principles from async materials



Discriminant Functions

- Define class boundaries, linear or nonlinear depending on the model

Gaussian Methods

- Use a probabilistic approach to classification

Perceptron & Logistic Regression

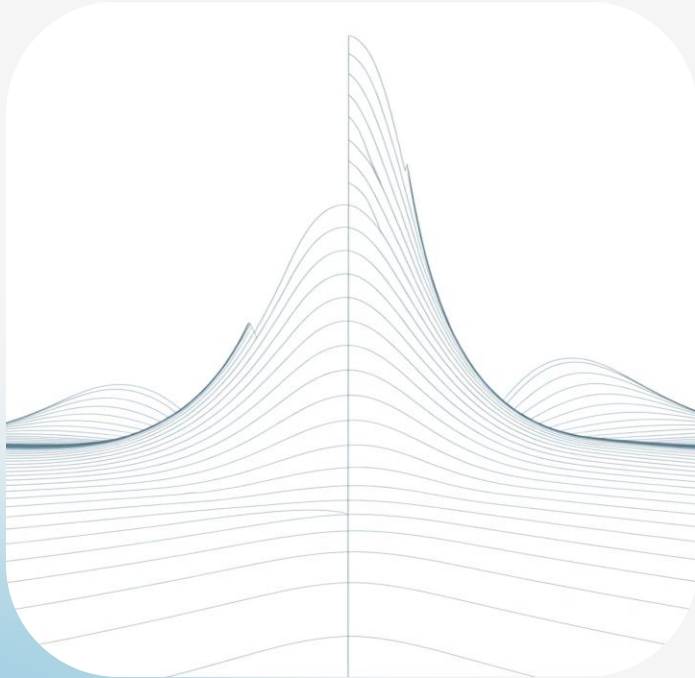
- Learn discriminants directly from data

GMM

- Create nonlinear boundaries using mixture models

Demo 1: Gaussian Contours on Churn Data

Visualising Class Overlap Using Gaussian Statistics



1

Compute Statistics: Find Gaussian mean and covariance for each churn class

2

Plot Contours: Visualise probability density for both classes

3

Analyse Overlap: Spot uncertainty regions between churn and non-churn

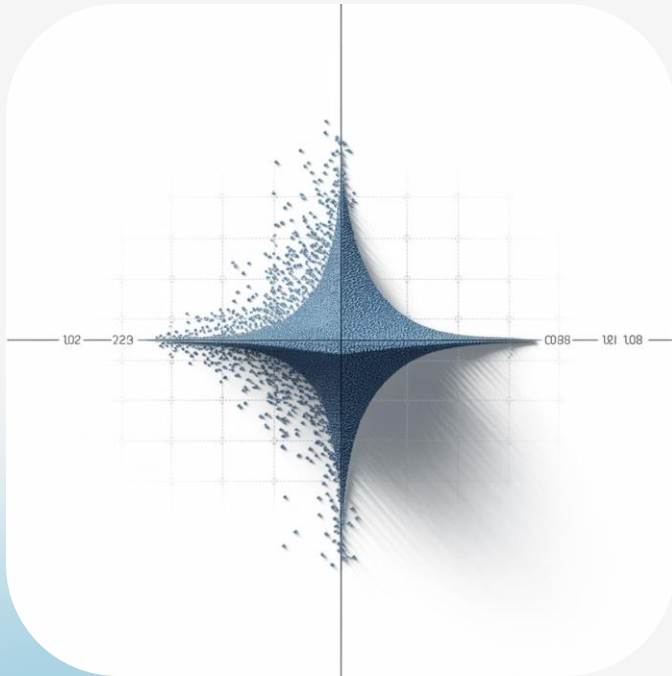


Key Observation:

Overlapping contours show uncertainty areas where churn and non-churn data mix, making boundaries less clear and predictions less certain

Demo 2: Predictive Maintenance

Linear vs Nonlinear Discriminants



Dataset Features

- Temperature readings
- Vibration levels
- Maintenance status

Models Compared

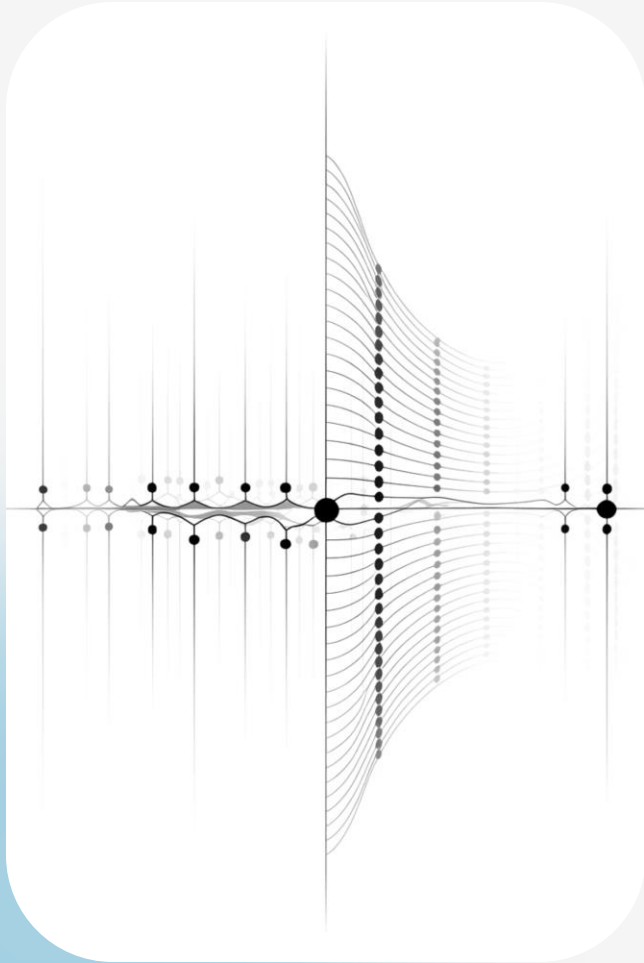
Logistic Regression (Linear)

KNN (Nonlinear)

- **Linear Models:** Create clear, simple boundaries
- **Nonlinear Models:** Capture complex, curved patterns

Demo 3: Perceptron on Churn Data

A Linear Classifier Demo



Objective: Trains a linear classifier to identify churning customers



Hard Linear Boundary: Creates a hard, sharp decision boundary



Feature Scaling: Sensitive to feature scales; requires data normalisation



Best Use Case: Best for data that is linearly separable

Key Takeaway: Learns direct boundaries without probabilities fast and simple, but only works with linearly separable data

Demo 4: Gaussian Mixture Model

Advanced Clustering for Customer Churn

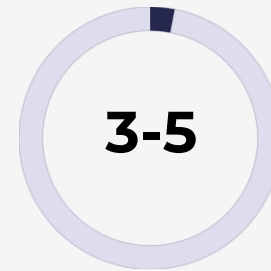
GMM (Gaussian Mixture Model)

- Combines multiple Gaussians per class to model complex data

Key Advantages:

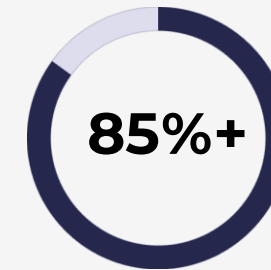
- Handles overlapping, multi-modal distributions
- Creates nonlinear, adaptive boundaries
- Captures multi-modal behaviour patterns

Real-World Application : Churn segmentation with varied customer behaviours



Typical Components

Number of Gaussians per class



Accuracy Gain

Improvement over single Gaussian on complex data

Recap and Reflection

Comparative Analysis of Discriminant Models

Model	Boundary Type	Learning Type	Data Fit	Interpretability
Gaussian Contours	Quadratic	Implicit	Simple overlap	High
Logistic Regression	Linear	Direct	Clean split	High
KNN	Nonlinear	Implicit	Flexible	Medium
Perceptron	Linear	Direct	Fast & simple	Medium
GMM	Nonlinear	Probabilistic	Complex overlap	Medium

Discussion

Choosing and Evaluating Discriminant Models



Linear Sufficiency: Is linear separation enough for your problem?

Boundary Curvature: How do covariance and mixture models change boundary flexibility?

Generalisation: Which discriminant models perform best on unseen data?

Next Step: Test these concepts on your datasets using different discriminant approaches

Q&A Session



Open floor for questions and clarifications.